Classification of Cancerous Cells

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STATS 101C: Introduction to Statistical Models and Data Mining

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December 15, 2025

I. Abstract

The goal of the paper and kaggle project is to predict the diagnosis status of lesions using statistical modeling. This report details the methodology (including exploratory data analysis, feature selection, feature engineering, data cleaning and model building) and results that our model achieved, its limitations and future work.

The final model is based on a general linear model which uses ridge regression using a dataset reduced to 19 final predictors that have been imputed with mean for numerical variables and keeps NAs in categorical variables as its own category "data not available". It has a kaggle score of 0.60750 and our final rank is 4th.

II. Introduction

In the United States, more people are diagnosed with skin cancer each year, roughly 5 million cases annually, than with all other cancers combined. Skin cancer occurs when DNA damage triggers the uncontrolled growth of skin cells, leading to malignant tumors in the skin's layers. It most commonly affects sun-exposed areas (Skin Cancer Foundation 2025).

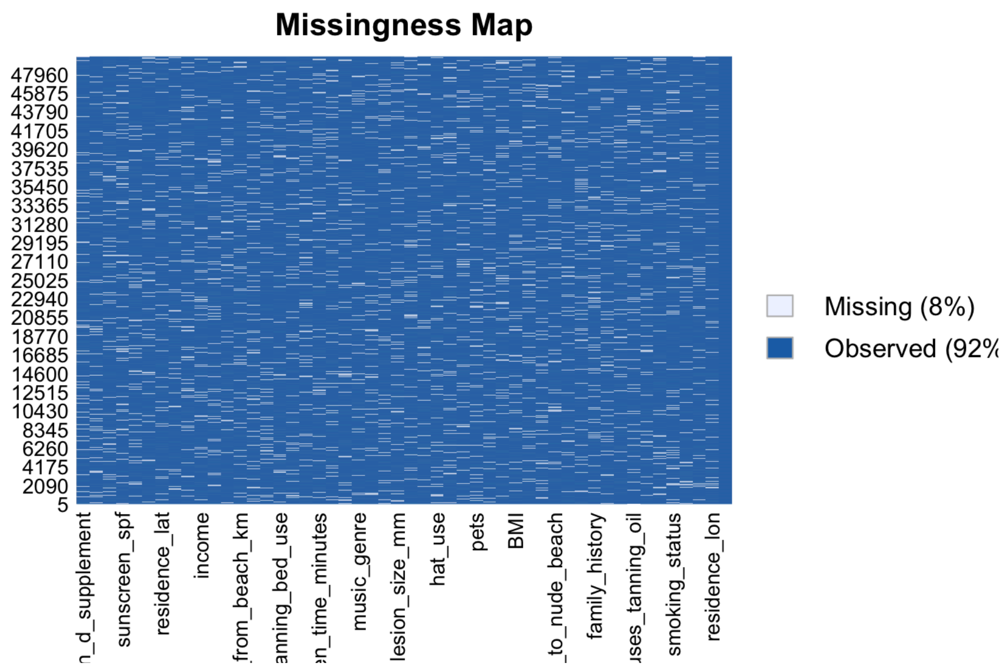
The prevalence of skin cancer is a serious concern, as more than two people die from the disease in the U.S. every hour (American Cancer Society 2025). Fortunately, many risk factors can be reduced, and early detection is highly effective. When caught early, melanoma (a cancer originating in pigment-producing cells called melanocytes) has a 99% 5-year survival rate (Skin Cancer Foundation 2025).

Current early detection methods primarily involve analyzing moles for irregularities in asymmetry, border, color, diameter, and evolution using the ABCDE rule (Aad 2025). This paper proposes using a machine learning model to predict the likelihood of skin cancer based on various factors from a dataset, determining if a lesion is Benign or Malignant. By providing an additional tool for early detection, this approach could improve treatment outcomes and patient survival rates. Furthermore, the model can identify the most significant factors contributing to melanoma, which may encourage preventive behaviors to mitigate the risk.

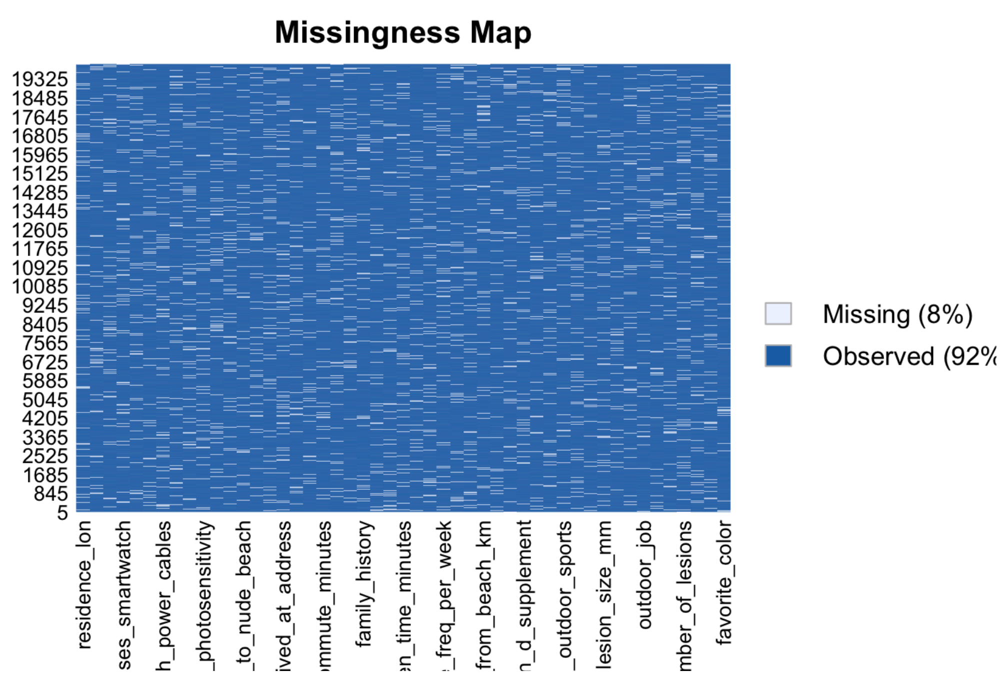
III. Exploratory Data Analysis and Variable Selection

Before working on the classification models, we did some exploratory data analysis to learn more about the data and use that to best understand which models would work well with our dataset models and features. First we found the dimensions – there are 51 columns and 50,000 rows in the training and 50 columns and 20,000 rows in the testing. The first column was just the index column in both datasets, so we removed those and ended up with 50 columns and 50,000 rows in the training and 49 columns and 20,000 rows in the testing.

Next, we looked at the number of missing values/ NAs in both the training and the testing dataset. There were a total of 196,042 missing values with an average of 4000 NAs in each column in the training dataset. 7.84% of the training dataset was missing, which is quite common for medical data as patients don’t want to answer many questions regarding their lifestyle.



Similarly for the testing data, 78,256 values were missing with an average of 1600 NAs in each column. Just like the training, 7.98% of the testing data was missing too.



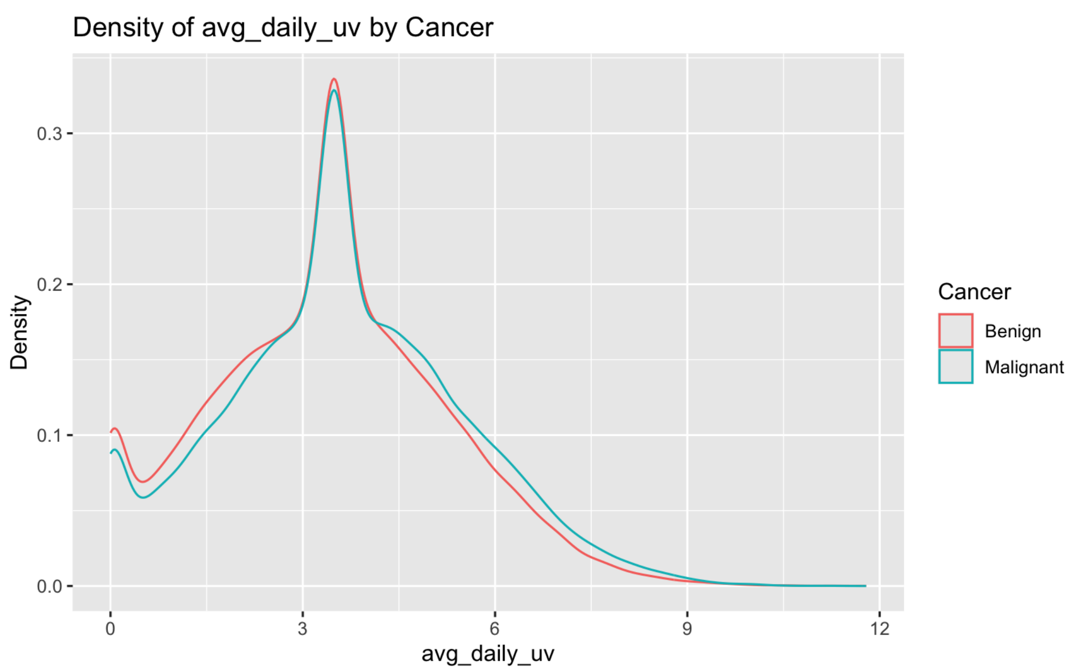
There are 29 categorical variables and 20 numeric variables.

To understand their impact on predicting whether a lesion is Malignant or Benign, we did some in-depth analysis of all the variables, split by categorical and numerical variables.

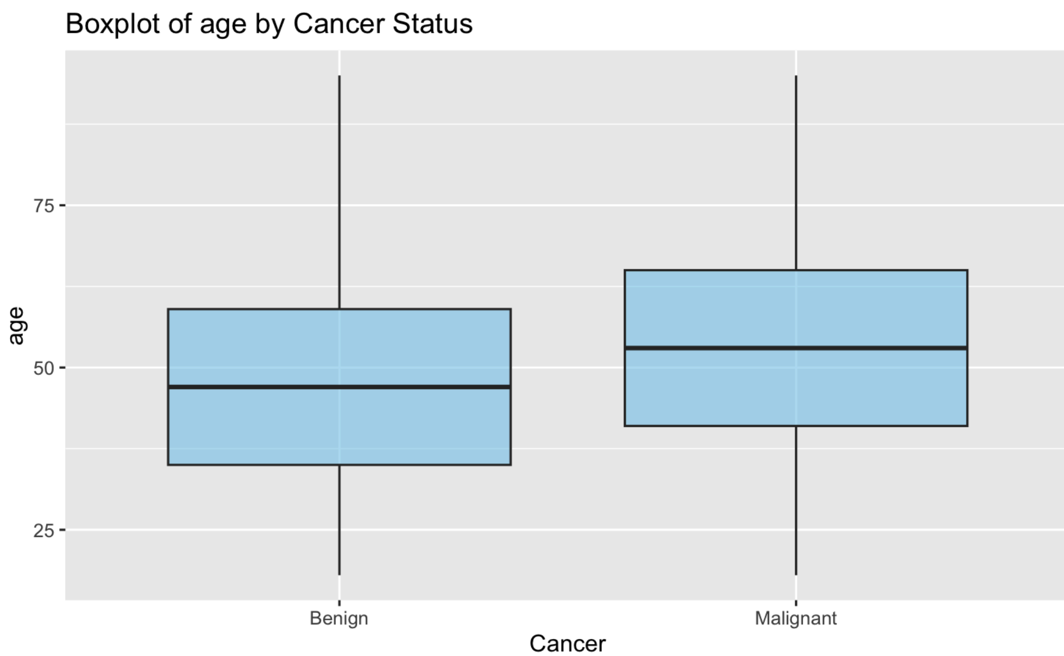
A. Numerical Variables

Some predictors like *BMI* and *Income* were inputted as numerical but their actual numerical value does not provide extra information beyond *underweight, normal, overweight and obese* for *BMI* and *low, middle and high* for *Income* so, we converted those to categorical variables with 4 categories each.

To find the best numerical predictors, we decided to make density plots or 'who's your daddy' plots. But they were not helpful in assessing which variables had significant differences between their occurrences for malignant and benign.

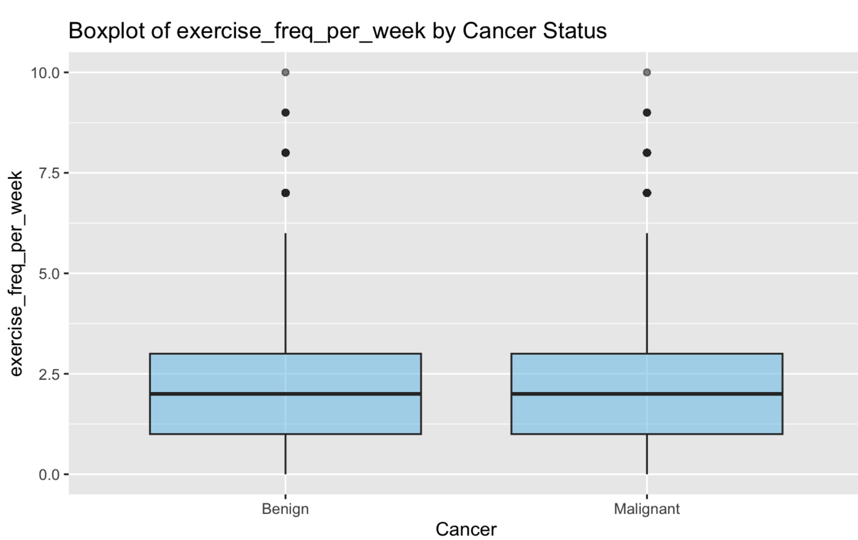


So we decided to compare the difference in medians between Malignant and Benign lesions for all the numerical predictors. We created boxplots to better visualize these differences.



Predictors like *age, avg_daily_uv, number_of_lesions, sunburns_last_year, lesion_size_mm, sunscreen_spf and years_lived_at_address* had significant visual differences in medians for malignant and benign lesions.

Whereas, predictors like *alcohol_drinks_per_week, frequency_doctor_visits_per_year, exercise_freq_per_week, commute_minutes, zip_code_last_digit, monthly_screen_time_minutes* and more did not have significant differences.



Further, to confirm our selection based on the boxplot, we also did a One-Way ANOVA test to compare numerical variables across *Cancer* levels. We extracted the p-values to test whether the numerical variable differs significantly between *Cancer* levels. The results are as follows:

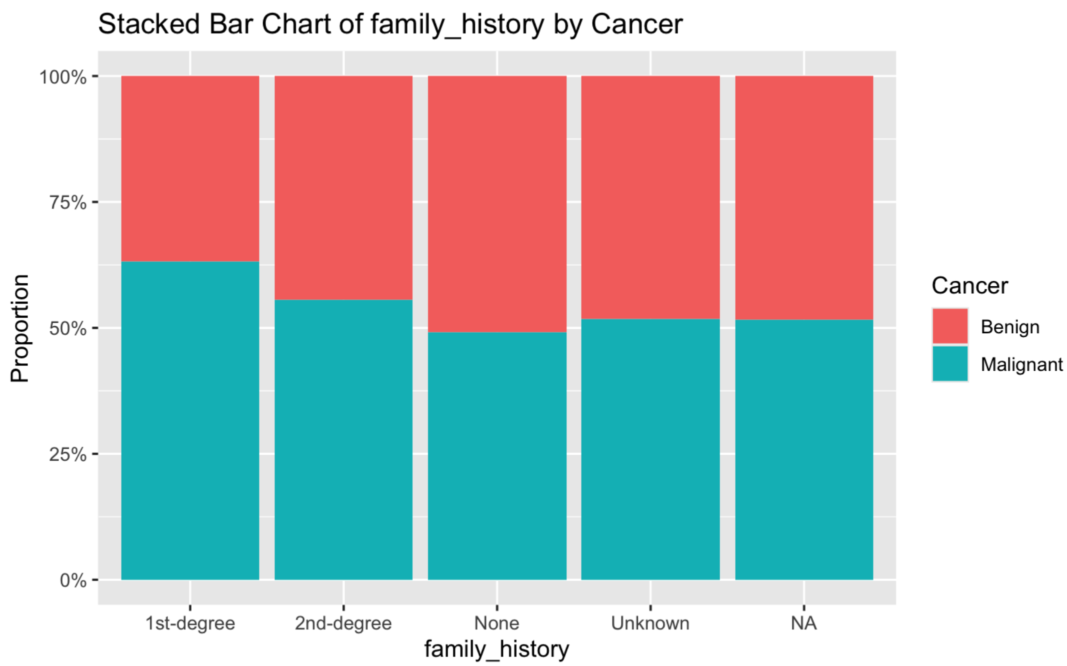
age.Pr(>F)1*	avg_daily_uv.Pr(>F)1*
1.022329e-203	5.377946e-41
number_of_lesions.Pr(>F)1*	sunburns_last_year.Pr(>F)1*
2.998532e-32	1.199340e-26
outdoor_job.Pr(>F)1*	lesion_size_mm.Pr(>F)1*
1.778298e-24	5.412025e-07
sunscreen_spf.Pr(>F)1*	years_lived_at_address.Pr(>F)1*
1.038599e-04	2.665885e-02
desk_height_cm.Pr(>F)1*	residence_lon.Pr(>F)1
6.473242e-02	1.079351e-01
exercise_freq_per_week.Pr(>F)1	zip_code_last_digit.Pr(>F)1
2.068466e-01	2.998131e-01
distance_from_beach_km.Pr(>F)1	alcohol_drinks_per_week.Pr(>F)1
4.021359e-01	5.972993e-01
commute_minutes.Pr(>F)1	residence_lat.Pr(>F)1
7.943421e-01	7.968784e-01
frequency_doctor_visits_per_year.Pr(>F)1	monthly_screen_time_minutes.Pr(>F)1
8.425452e-01	8.943188e-01

Looking at this, we also added *desk_height_cm*. While desk_height_cm does not seem medically relevant it had a significant difference in median and later impacted the prediction accuracy of the model.

Thus, we ended up with 9 numerical predictors from the data analysis.

B. Categorical Predictors

For categorical predictors, we decided to make stacked bar charts to compare their proportion differences.



Variables like *skin_tone, occupation, sunscreen_freq, hat_use, clothing_protection, tanning_bed_use, family_history, immunosuppressed, and skin_photosensitivity* had visually different proportions of malignant and benign in different levels of the categorical predictors.

Below is a table containing the number of levels in each categorical variable (excluding NA).

Predictor	Number of Levels
skin_tone	6
occupation	7
sunscreen_freq	5
hat_use	4
clothing_protection	3
tanning_bed_use	4
family_history	4
immunosuppressed	2
skin_photosensitivity	3

Below is a table of the top results from the chi-squared test between the response and each predictor to recheck the results from the preliminary analysis.

family_history*	***skin_tone***	***sunscreen_freq****
9.781167e-118	*2.951334e-81*	*9.786698e-49*
immunosuppressed*	***occupation****	***tanning_bed_use****
4.517045e-41	*1.241510e-19*	*1.732939e-18*
clothing_protection*	***skin_photosensitivity****	***hat_use****
6.437897e-15	*1.053908e-12*	*5.860108e-10*
urban_rural	*income*	*participates_outdoor_sports*
1.015537e-01	*1.110831e-01*	*1.753640e-01*
uses_smartwatch	*music_genre*	*vitamin_d_supplement*
2.096736e-01	*2.372982e-01*	*2.551280e-01*

The test confirms the predictors we chose from the stacked bar charts. Thus, we have 9 categorical predictors.

From the exploratory data analysis, we ended up with 18 total predictors – 9 numerical and 9 categorical predictors.

IV. Methods and Models

A. Methods of Imputation

Our main method for evaluating the models used was cross-validation, either using k-fold cross-validation or Leave One Out Cross Validation (LOOCV), rather than a training-validation split. We wanted to ensure that we reduced variance and got a more accurate measure of the model. The test data scores from Kaggle were more similar to the accuracy scores we gained from cross-validation. We also utilized various imputation methods, such as median-mode, mean-mode, MissForest and adding a separate *missing* level for categorical variables-mean for numerical. Various models tried were Logistic Regression, LASSO, Random Forest, and Linear Discriminant Analysis (LDA).

a. Median

To handle missing numeric values, we used the first and most standard measure of center on the numerical variables – the median. The median is often considered the safest measure of spread as it is the most resistant to asymmetric distributions and extreme outliers. Thus, with so many predictors, we relied on the median as an initial imputation method. However, it was discovered that other methodologies performed better.

b. Mean

While both mean and median are basic imputation techniques, we tried mean imputation for numerical variables as well to be able to train the model better, accounting for outliers presence. The models with mean imputation performed better than the median imputation and in our final model, we used mean imputation with scaling for numerical variables.

c. MissForest

We also considered MissForest imputation as an imputation method. This method first imputes the missing values with Mean-Mode imputation, then utilizes the observed values as the response values and the rest as the predicted values to train a Random Forest model. The model is then used to impute the missing values. This imputation method performed second best overall, with a Kaggle score of 0.60415 when used on logistic regression. However, we found a more effective imputation method later.

d. Mode

Mode was initially used to impute the categorical variables. The expectation from this imputation method was that if a specific category has a higher percentage of data points that is said category is the most representative for the column and that category should be imputed for the missing values. However, this seemed to decrease our accuracy.

e. Separate *Missing* Category (categorical variables)

The best-performing imputation method then was to use "data not available" in place of the missing values, making it its own category. By keeping them as missing, we were able to maintain the pattern and use that missing data as information to increase our model accuracy. This was also the best choice because ~7% of the entire data set was missing, indicating that the mode as an imputation method may not provide an accurate insight into which category is most representative of the column. This imputation method gave us our highest score (0.6075).

B. Models

a. Logistic Regression

The first model that we decided to use was logistic regression, as it is not complex and highly interpretable. This was initially run on all given variables to assist with initial variable selection to identify significant predictors, as well as on our selected predictors. With this model, we were able to get an overall score of 0.60475 on kaggle when run on all of our predictors using median-mode imputation. However, accuracy decreased when fitted to the subset of predictors, indicating one of the pitfalls of logistic regression, as the number of predictors goes up, the chances of overfitting get higher. Complexity in models can lead to an increase in variance as bias goes down, and so it does not bode well for its ability to fit to new data, making logistic regression unreliable despite its high score on the testing data.

b. Linear Discriminant Analysis (LDA)

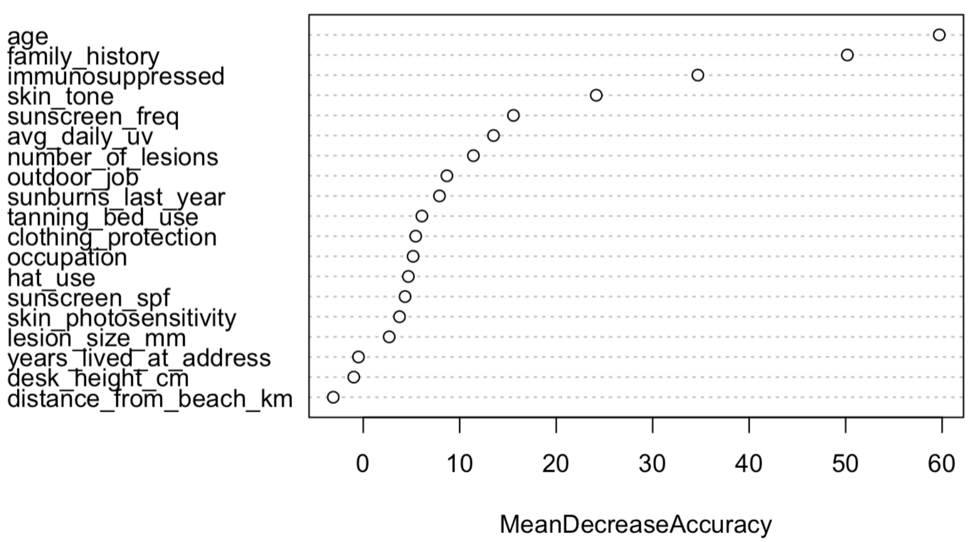
Linear Discriminant Analysis, or LDA, is a classification algorithm that relies on the Bayes classifier and attempts to find the linear combination of features that separates the two classes. It relies heavily on the predictor covariance being equal within each class. It can also be used for dimension reduction purposes, which was attempted, but overall made the accuracy drop too much when applied to be considered. LDA performed best with MissForest classification on the full dataset, with a Kaggle score of 0.6036. It also outperformed QDA, which had a high score of 0.5756, revealing that the boundary between the two classes was definitely linear. This model was again too complex, giving too high a variance to be reliable.

c. LASSO

Logistic Regression with LASSO is a regularized classification method that aggressively shrinks coefficients to 0, and thus is ideal for reducing model complexity and determining which variables are significant. This was used both as an initial model on all predictors as well as on our chosen group of predictors in the hope of performing variable selection. However, the best performing LASSO model gave an accuracy of .6004, and had a lambda of 0.0005, indicating that variable selection did not occur, as well as a decrease in accuracy from logistic regression, which was able to perform more variable selection via significance.

d. Random Forest

Random Forest is an ensemble classifier made of decision trees that runs very efficiently on large datasets, making it an attractive option for this cancer dataset. Using cross-validation, we were able to determine the best model to be features sampled per split to be 7 and 1000 trees, giving us a Kaggle score of 0.5963.



We used the Variable Importance Plot to identify the best subset of variables, which were *age, family_history, immunosuppressed, skin_tone, and sunscreen_freq*, and then ran a new model with them. This, unfortunately, decreased training accuracy from 0.5847 to 0.5842, further confirming our variable selection as the correct balance between complexity and accuracy. Random Forest was unable to outperform the previous models, even with hyperparameter tuning.

e. Generalized Linear Model (GLM) (Best Model)

From the perspective of multiple regression analysis, a generalized linear model (GLM) predicts the variation of a dependent variable in terms of a linear combination of several reference functions. Because of its flexibility to incorporate multiple quantitative and qualitative independent variables, the GLM has become a powerful algorithm.

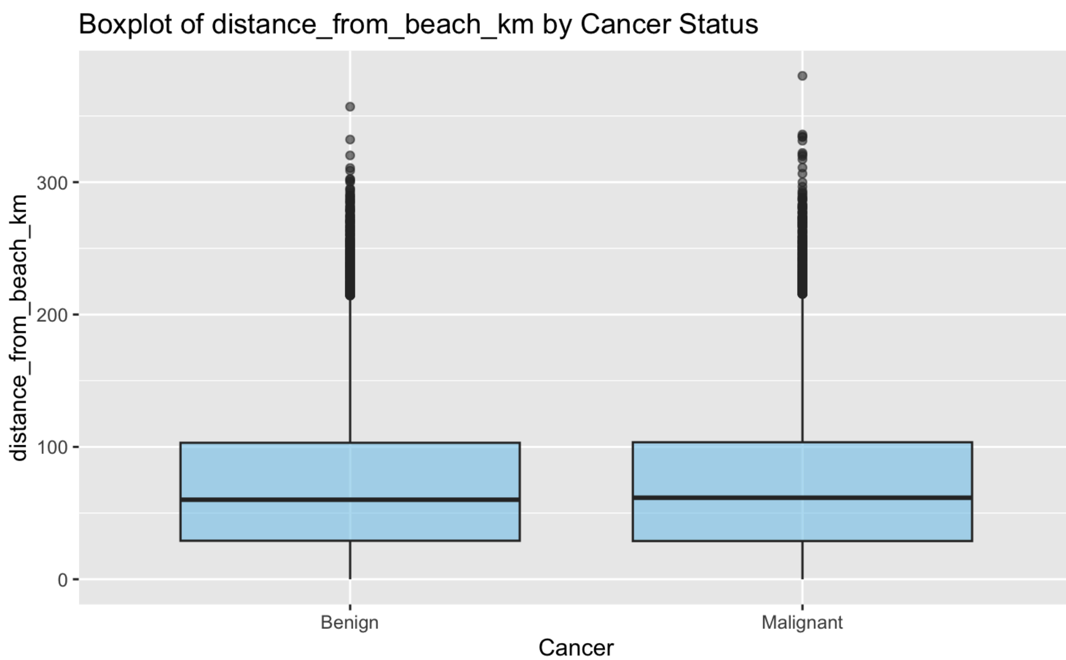
For imputing missing values in numerical variables, the mean was selected. In addressing the issue of missing values in categorical variables, we conceptualized these as a distinct category, "data not available." As previously indicated, missing values accounted for over 7% of the data, constituting a significant portion.

This decision was also made considering that this dataset is medical data. We fitted a regularized logistic regression model using elastic net penalty ($\alpha = 0.5$), which combines ridge (L2) and lasso (L1) regularization. In the context of medical data, aggressive variable elimination may result in the discarding of clinically relevant information. The ridge component of the elastic net allows all predictors to remain in the model while simultaneously controlling their magnitude. This approach serves to mitigate the occurrence of overfitting without forcing coefficients exactly to zero.

When running this model on all predictor variables, the Kaggle score was 0.60480. This model demonstrated the best performance compared to other models, and further tuning was performed to align with our chosen group of predictors.

To further maximize model accuracy, we investigated the remaining predictors excluded from the final list of 18. Given that mean imputation produced the most significant results, we re-examined the numerical variables.

The boxplot for distance_from_beach_km suggests its inclusion might improve the accuracy score, as the median distance differs noticeably between Benign and Malignant cases.



We ran the model with the selected 18 predictors, as well as an additional version that incorporated distance_from_beach_km. The best performance was attained when utilizing a total of 19 predictors. After we submitted our prediction on kaggle, the accuracy was 0.60750, which is the highest score.

V. Discussion and Limitation

Despite placing 4th on the overall leaderboard, we didn't stop testing different approaches to achieve a higher accuracy score, as we found 0.6075 to be unsatisfactory. There were several factors that impeded our attempts to raise our score, as well as limitations we found in the models we did use.

We observed that all of our top-performing models yielded similar accuracy scores, with our highest scores hovering around 0.60 even with different imputation methods. This suggests that another classification model may be necessary to break the current performance ceiling.

Determining the best imputation method for the missing values in our data also presented a significant challenge. We first tested standard imputation methods such as Median-Mode and MissForest, which performed reasonably well: our Logistic Regression model with Median-Mode imputation gave us an accuracy score of 0.60475. However, we actually achieved our highest score upon treating "missing" values as its own category. This implies that the missing data may not be missing completely at random. Thus, further insight on the medical information of patients with missing metrics would be rather helpful towards building a more accurate model.

Another limitation we encountered was model assumptions. Our top models, GLM and LDA, rely on structural assumptions such as linearity or equal covariance. Such assumptions downplay the complexity of biological processes that contribute to the development of skin cancer. The performance ceiling of our models also suggests that linear classifiers are insufficient to capture the complex nature of skin cancer diagnosis.

Finally, we found that imputing and modeling on 50,000 rows and up to 50 predictors is computationally expensive. This proved to be a constraint in our work because it was rather slow, giving us less opportunity to fine-tune our models within our limited timeframe.

VI. Conclusion and Recommendation

After extensively evaluating several classification and imputation methods, our most effective model turned out to be GLM with ridge, with treating "missing" as its own category. This approach yielded an accuracy score of 0.6075. While other attempts using LDA models resulted in similar scores, GLM is still preferable due to its simplicity and interpretability.

However, the several limitations we encountered in our process prompted us to consider future steps to overcome the apparent performance ceiling. Since model assumptions such as linearity appeared to contribute greatly to our scores plateauing, future work would explore more complex, non-linear models such as Artificial Neural Networks. We would also like to further investigate the nature and causes of missing data, such as patient demographics or medical history, to better understand potential biases in the data, as well as the best method to impute missing values in the context of this dataset.

VII. Acknowledgement

We express our gratitude to Professor Akram Almohalwas for his extensive guidance in Stats 101C this fall. We are greatly appreciative of his constructive feedback and support, which have contributed to the development of our best work.

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